

What are people's limits when it comes to generative AI in video games?

Khang Nguyen - S4005711

Jimmy Vu - S4006701

Mitchell Long – S4007880

A look into how gamers react to the use of generative AI in video games.

GitHub Repository: <https://github.com/MitchKLong/COSC3047-Assignment-2>

Introduction and Background

Generative AI has been a hot topic in the last half-decade, being used in many different industries, but for this report we will focus on its usage within game development. Using generative AI, game studios can utilise these tools to almost instantaneously generate whatever they want, whether it be textures, models, music and art.

Recently, the game “Clair Obscur: Expedition 33”, which took the mainstream gaming space by storm in 2025, has sparked controversy after admitting to using generative AI assets early on during its development, with many players having mixed feelings due to the usage of it not being fully reflected in the final release, but the fact that it was still used at all has shown to be a topic of contention. The CEO of the SHIFT UP corporation known for game development and publishing has also gone under flak and backlash due to their support for AI stating it as an essential tool being able to allow one person to perform the work of 100 people.

Another instance of AI being used in game development was Battlefield 6, the latest entry in the popular FPS series. Around December of 2025, many players were quick to notice one of the in-game cosmetics is seemingly generated by AI. This was a major controversy as Electronic Arts (EA) made statements prior that no generative AI content would be featured in the final product.

With these few examples of generative AI in video games, we aim to use our research to analyse the negative discourse and community surrounding the topic of generative AI in game development, how more specifically how people tolerant people are to having it be used, while seeing if there are any nuances to the topic.

When we were deciding which platform to utilise to explore our question, we opted to use BlueSky for its ease of access to its API, which requires no API keys, having no daily limits to fetching or reading posts, requiring only an account, which we made for the purpose of this assessment.

Data Collection

With Python, we created a script that gathers posts from Bluesky using search queries. The queries we have set up are to help narrow down the topics of gen ai towards games to help with finding post discussing and exploring our topic. We decided to not limit as to how old the posts are as this is still a recent topic and majority of the queries that are focused on games are curated for recent incidents.

The queries we used were: "Gen AI, AI, games", "Gen AI, AI, Larian", "Gen AI, AI, Expedition 33", "Gen AI, AI, Battlefield 6", "Gen AI, AI, Shift Up", "Generative AI, Games", "Generative AI, Games, Good", "Generative AI, Games, Bad" and "AI Slop, Games".

There was however a limitation to using the BlueSky API; we were only able to call 100 posts at a time. We were able to circumvent this limitation by looping through an array storing the queries we have created. This way we can also get a broader result from using the different queries and get a larger dataset.

In total, due to the limitations of the API, we were to collect about 732 posts along with their comments; the total amount of comments was 783.

Part A - Network Connections

This part of the report aims to explore the social network and community surrounding discussions related to generative ai in games. Spotting any popular figureheads or users involved within the discourse and displaying their first-degree connections and replies to see how much outreach they possibly can output to their communities.

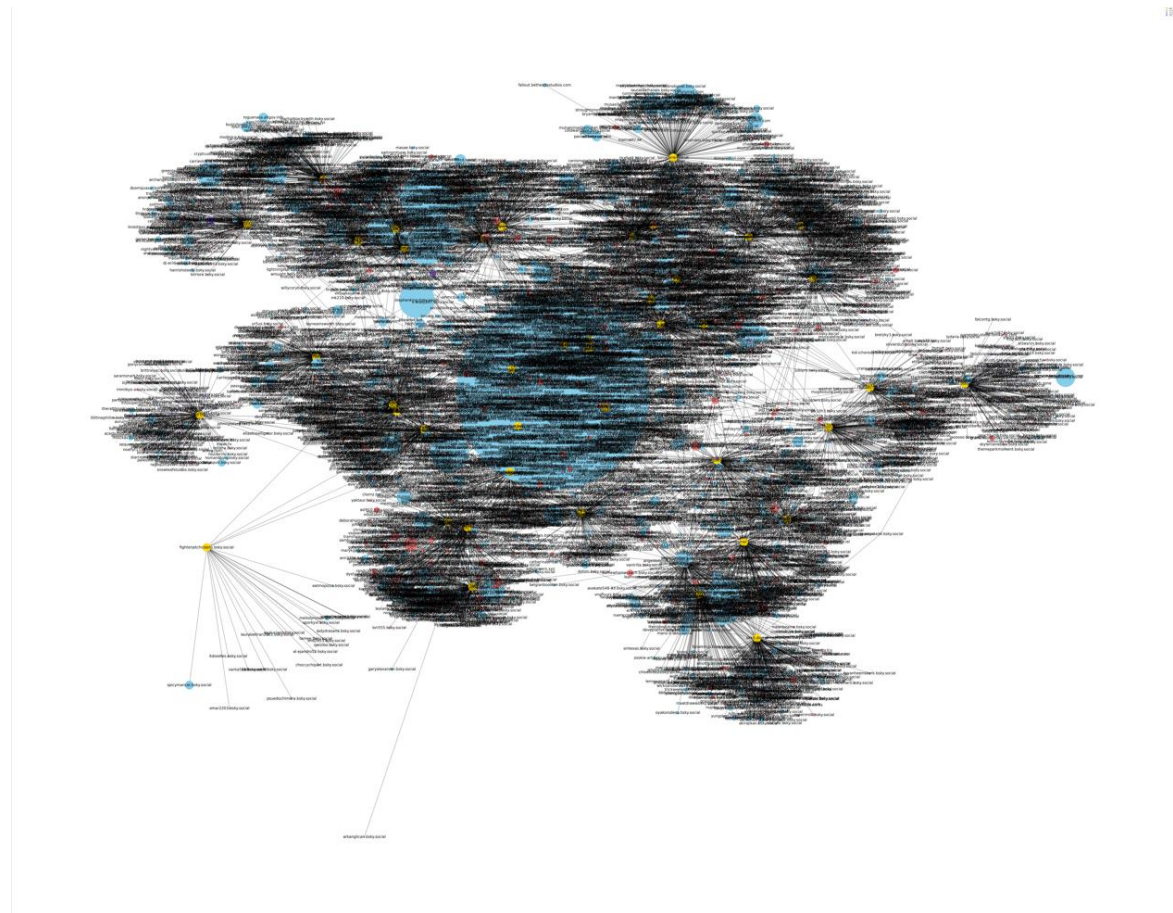
Visualising the Ego Network

After collecting all the data required. We decided to run network analysis on the most influential posts found from the queries by sorting through the posts with the most liked. We then took the top 50 post authors to collect their first-degree followers and following.

After initialising a digraph, each post author of the top 50 was added as ego node and their respective followers and following list were each added as separate nodes with directed connected edges with the author. Their follower counts were also collected appended to denote node value.

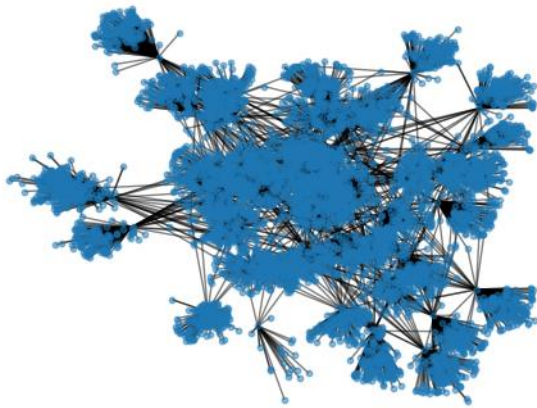
Unfortunately, Bluesky employs a limit of 100 users per collection and even then, running Api requests for each user profile takes a huge amount of time to run (40+ minutes) for 50 post authors. We decided this was found to be enough due to the cross connections between the 50 ego nodes which shall be shown through our following graphs.

First, we will be showing the graph we have displayed with networkx. The ego nodes are set as gold, followers are then shown as red with the label out-degree meaning the ego influences out to those users whereas the users following the ego are shown as blue meaning they influence the ego with the label in-degree.

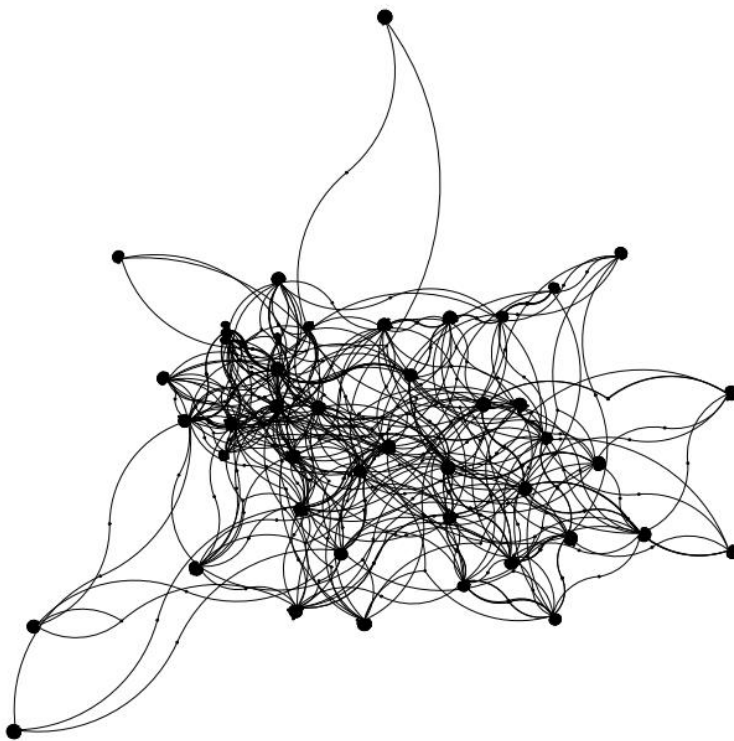


As we see from the graph that there are separate isolated components related to each ego with the egos in the centre being shown to follow users with large follower counts displayed by the large size of the blue nodes.

We also attempted to use a Kawada Kawai layout graph, but this became highly limited due to the outright size of the graph, as you can see below it comes off more difficult to interpret the results but still displays segregated sections for each ego. We had to turn off arrows and labels to allow visibility, and with a node_size to 10 as the default 300 size came off as too large.



Now we can show a graph we have generated using the Gephi software, this uses the ForceAtlas 2 layout displaying connections amongst all the nodes.

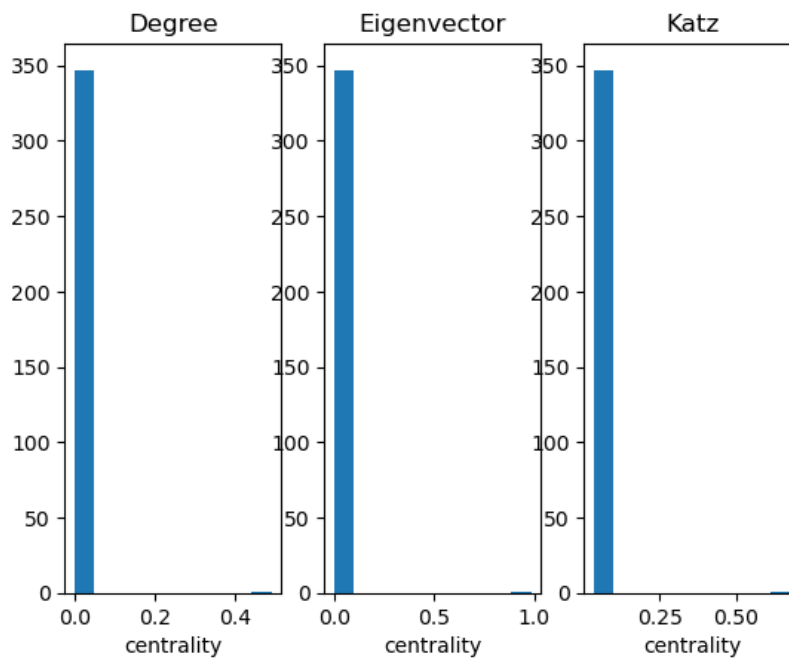


We can conclude from the 3 graphs that despite each ego having their own separated section of users they follow and are following. There are still connections between the first-degree connections of ego nodes with other ego nodes, displaying that the discussion of generative ai in games in Bluesky is very intertwined as many first-degree connections tend to connect with other first-degree connections meaning that the influence of ego nodes is very high and that there consists of second-degree connections between egos.

SNA Measures and Centrality

Now for the rest of **part A**, we will be using a reply graph of all the collected posts with edges directed towards the replies of each post. We have then omitted any posts that do not contain any replies. Unfortunately due to limitations, this graph also only contains first-degree replies due to the limit of the Bluesky API only collecting first degree replies and if we wanted to collect more than the first degree, we would have to loop through each reply unknowing if there are every any replies. This would require large amounts of time to be done.

For calculating centrality, the reply graph was run through degree, eigenvector and Katz computations. This was done with the networkx library and plotted with the matplotlib.pyplot library.



Overall, centrality is shown to be overall very low for all computation methods. We believe this may be due to the nature of the data, as we only have first-degree replies meaning we are unable to see if there are any proper reply chains with a commenter and a post author which are a huge part of centrality. The small, tiny parts of upped centrality near the end the centrality axis we believe is due to the post authors as they can still have multiple replies albeit separate users so they will tend to have more edge connections to any user that has only replied.

Below we have 3 Gephi graphs with each computation method in the respective order of degree, eigenvector and Katz. This is done with the Yifan Hu Proportional layout.

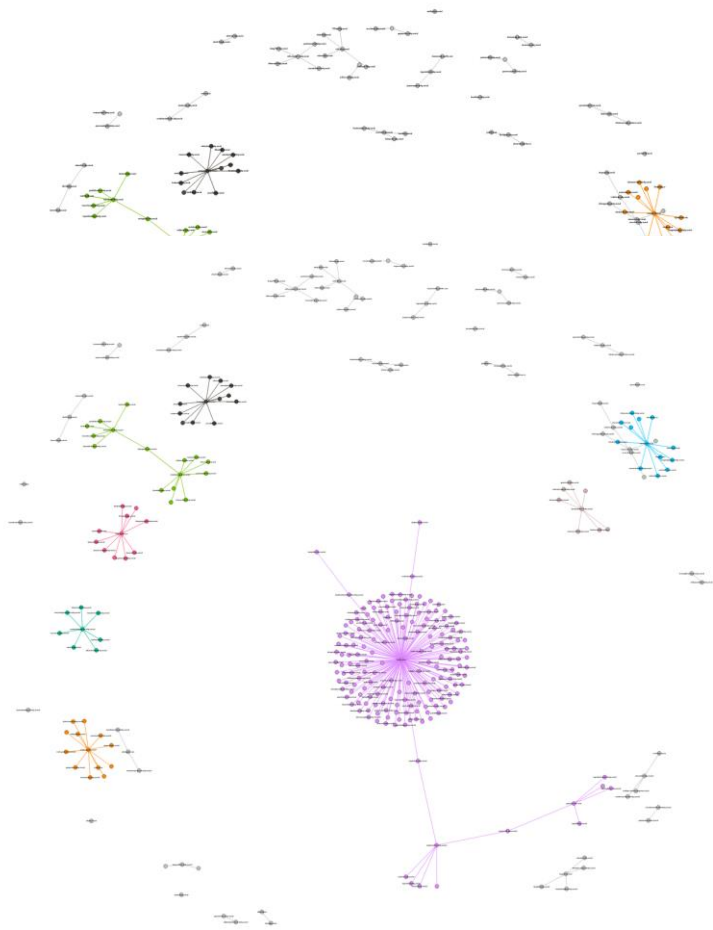


Each computation displays similar results due to all having a similar amount of centrality.

Community Detection

Clique Percolation Method (CPM) is a community detection algorithm that works by finding overlapping cliques in the graph. It uses a k -value to find cliques of k -size and if they share enough overlapping nodes, it merges into one community. There was an issue however initially when we used a K -value of 3, but we were unable to get any results, until we lowered it down to 2. Even with the k -value lowered, it could not detect as many communities as compared to the Louvain method, where CPM detected 32 but Louvain detected 44.

Unlike the workshop for Week 9, we did not use a ground truth labels file. We found that it would take too much time to come up with ways to separate these individuals but also it would introduce human bias if we were able to sort them out ourselves if we were to write out definitions of how to detect a community. We want the algorithm to detect natural interaction within the social network and how people are connected in the discussion.



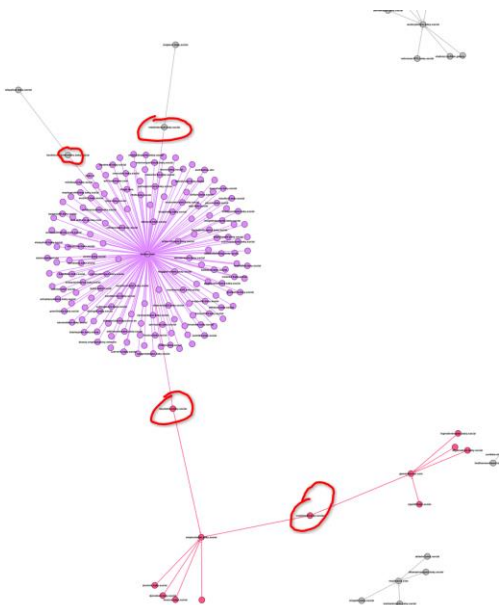
Using Gephi, the graph represented here is the reply graph detected using the Louvain algorithm. Represented using Yifan Hu Proportional.

Using Gephi, the graph represented here is the reply graph detected using the CPM algorithm. Represented using Yifan Hu Proportional.

Community Graph – Analysis

Both graphs are able to see similar communities, however for the largest community as seen in the centre, CPM detected all nodes connected to it as one big community, rather than being able to detect that it is a separate

community like Louvain did. Those communities that CPM could not detect were users who appeared multiple times in different comment sections within the posts we have gathered.

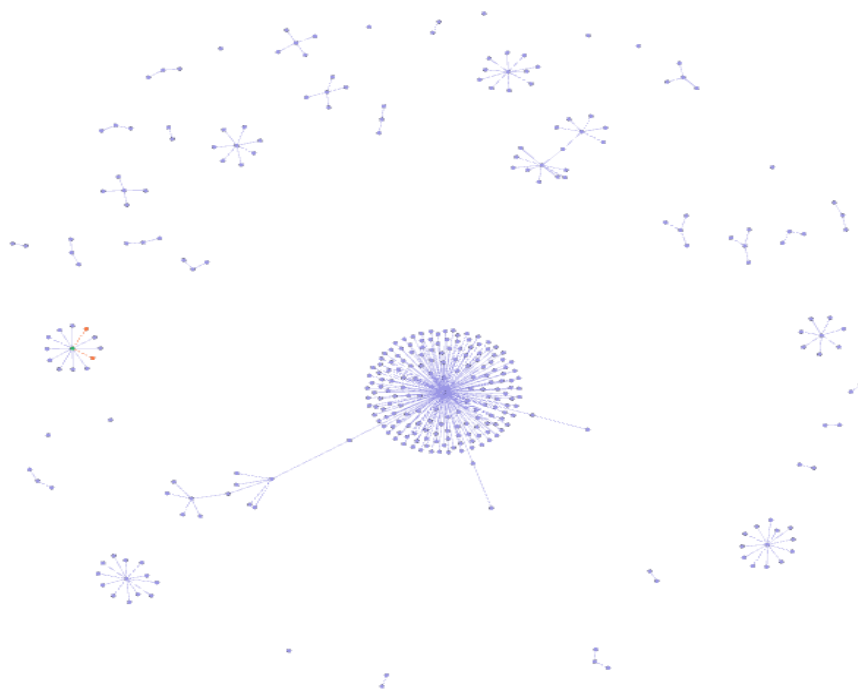


Highlighting these key individuals here, we see that there is a subset of individuals commenters who have an edge towards multiple nodes, which indicates that are people out there who are highly engage in discussing about the topic of generative AI in video games from different perspectives and posts. Unlike most users who replied to a single post, these users represent a group of individuals who like to engage in multiple perspectives. On top this observation we see that the graphs are disjointed with very little overlap between cluster,

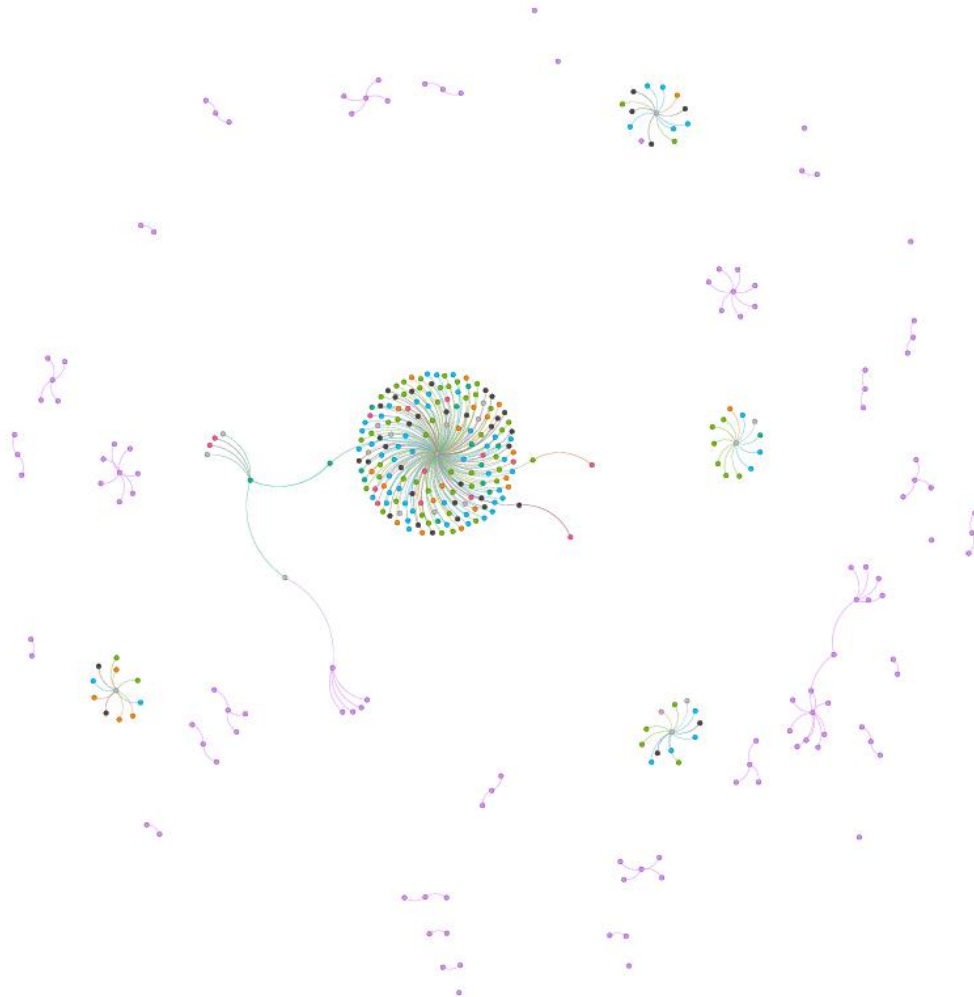
Independent Cascade

Originally after running the reply graph through independent cascade, collecting the Average Activations Per Node. We found that due to the first-degree reply graph nature of the data, the average activated nodes per cascade simulation were shown to be incredibly low. The only nodes that were shown to be activated were the initial 2 set in the seed to be cascaded upon. Below we have a graph generated with the Yifan Hu Proportional layout with the average activations partitioned.

Most nodes displayed the same colour with very few nodes displaying the other colours displaying a higher degree of activation.



Afterwards we had figured we may be able to create some interesting results if we converted the graph to be undirected and introduced a different cascade configuration for the seeded nodes. Below we have another graph generated with Gephi but this time showing a much more diverse set of colour output denoted the average amount of nodes activated largely increased through each cascade run.



Here we can see that the more isolated grouped components still display as purple as they are much smaller meaning than the main components denoting lesser activations due to the lower number of connections or interactions. With bigger components we can see a much larger variety of displayed colours meaning they have cascaded much further with a higher average activation due to the larger number of connections and interactions. This is most evident in the largest component in the middle.

Part B

In this part, we aim to determine the sentiment about generative AI and the different subtopics within the dataset. This will be done by apply three different sentiment detection models Vadar and two different BERT models to help us not only determine but also verify the sentiment. We will then apply topic analysis using Latent Dirichlet allocation to help us determine exactly what the people within the dataset thinking about beyond just positive or

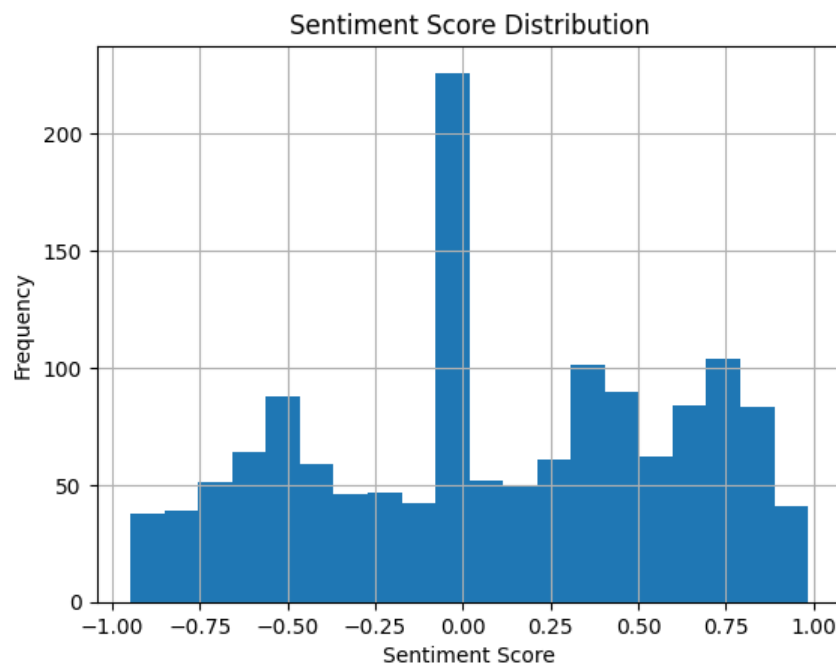
negative sentiment. Due to the reception to generative AI across the internet, our hypothesis is that there will be a negative sentiment towards generative AI in video games.

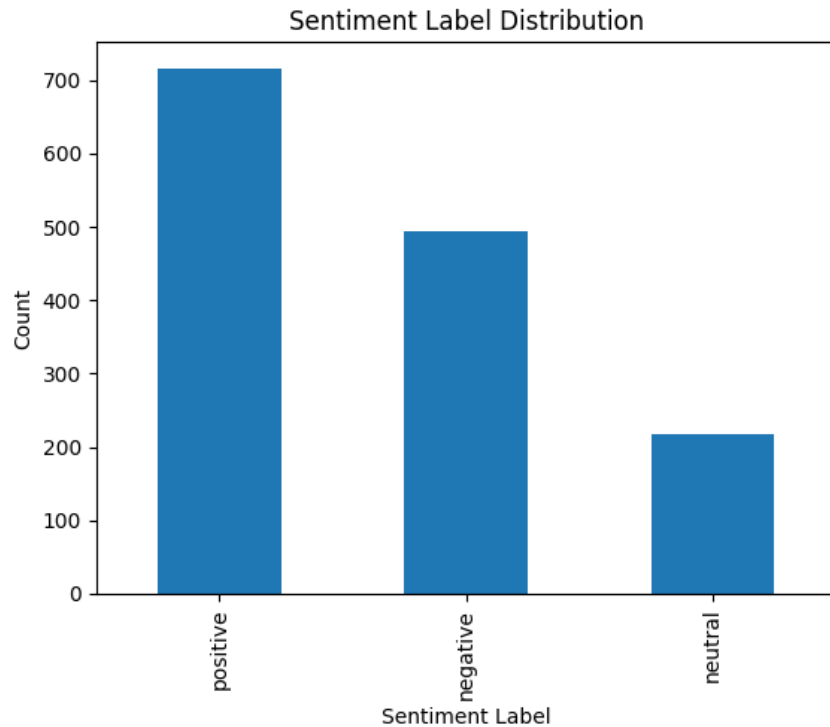
Part B.1 - Processing the dataset

Before we can pass the data into the different models, it needs to be processed so that any junk data is removed and the data we do care about is in a format that the models can understand. To do this, all text within the dataset is converted to lowercase, any URLs are removed because they are a common part of Bluesky comments but are ultimately useless for our needs. Any emojis are converted to their equivalent words and any non-letter characters are removed, the words are converted to tokens, the stopwords are removed and lemmatization is performed to convert the word into their most basic form. Finally, any negations are handled to ensure that the models get the correct meaning from them. Now the dataset is ready to be used to determine the sentiment.

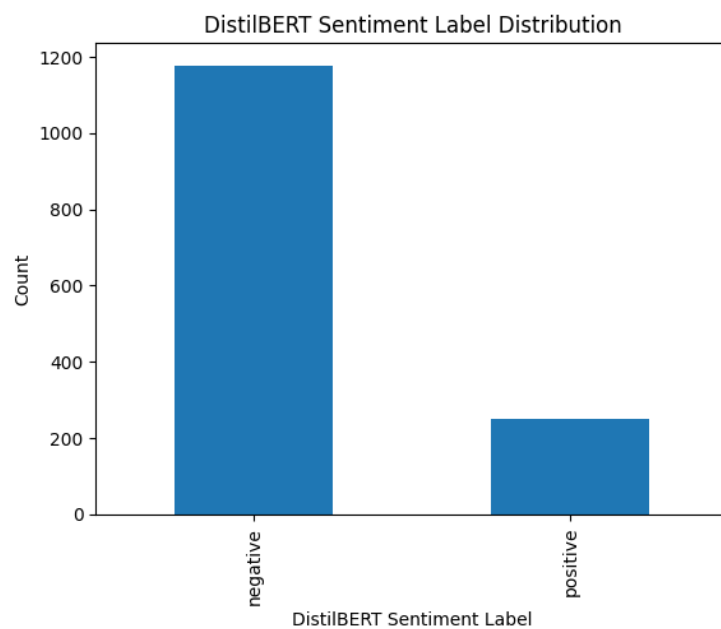
Part B.2 - Sentiment Analysis

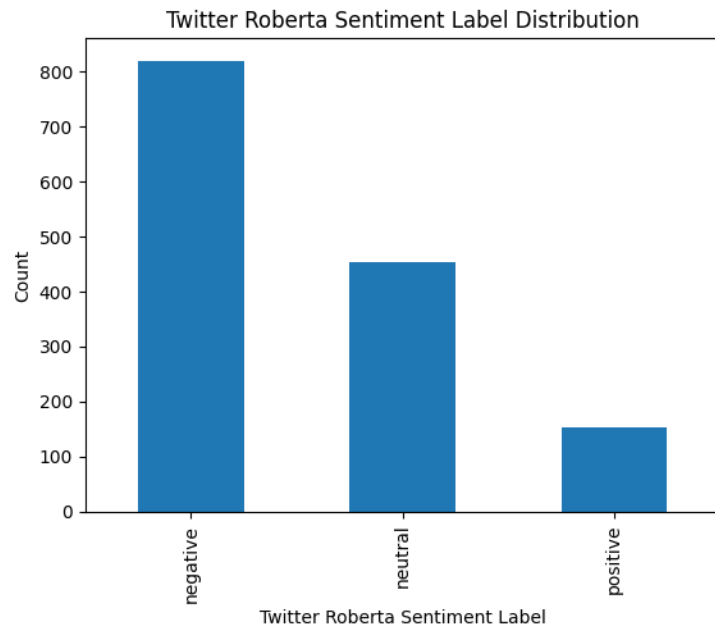
Now it's time to determine the sentiment of the use of gen ai in games. To do this we will first run it through Vader to get an initial result. This is shown in the histogram and bar chart below.





Both the bar chart and the histogram show that there is quite a lot of "neutral" values and there are more "positive" values than "negative" values. This directly contradicts our hypothesis so we will attempt to confirm or deny these results using two different BERT models. These will be Distilbert Base and Twitter Roberta Base. The results are shown below.

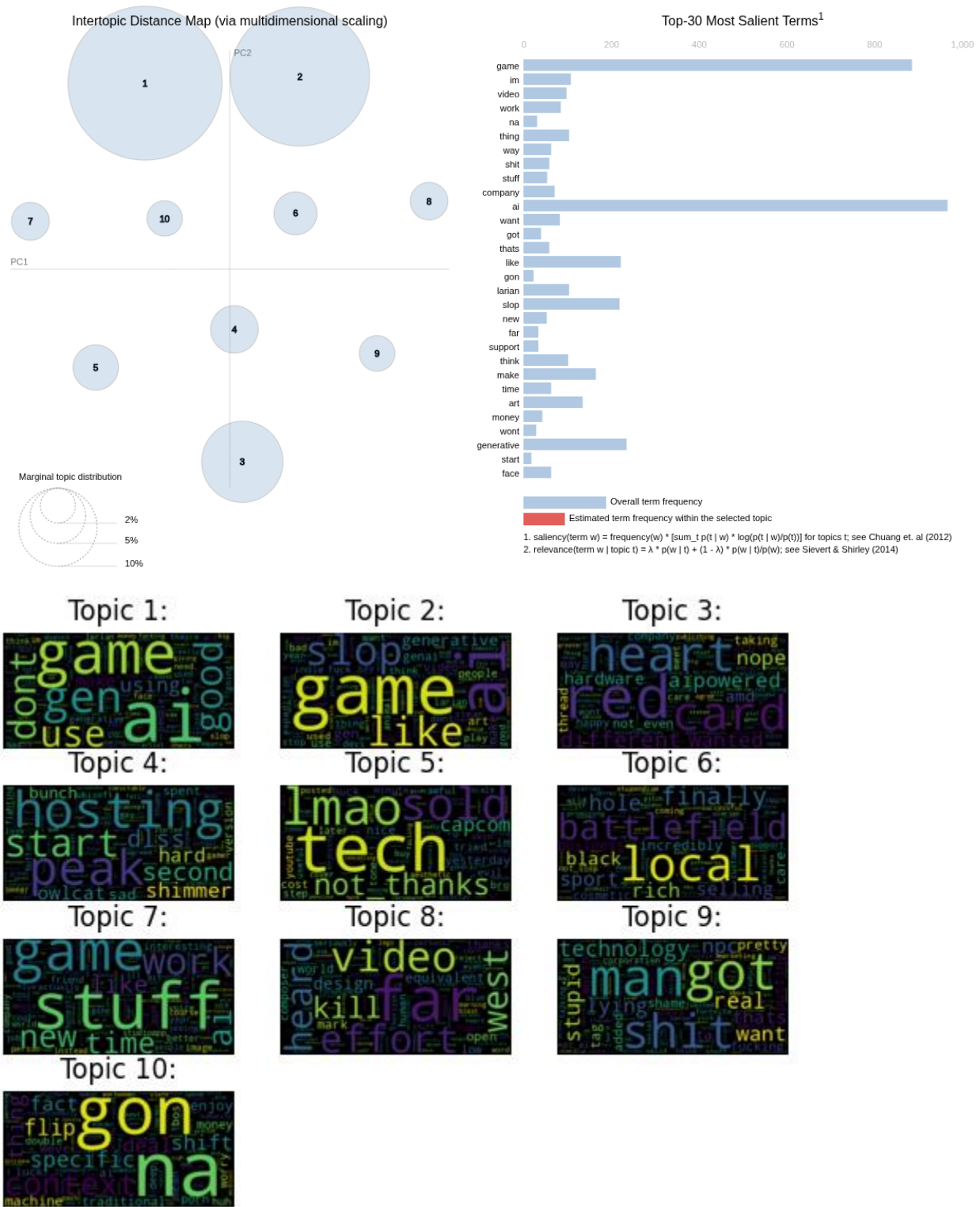




The results from both the Distilbert Base and the Twitter Roberta Base models clearly show that our initial assumption is correct unlike the Vader model. This could be because of several factors, but it is most likely due to the major difference in the models. Due to the correlation of results between Distilbert Base and Twitter Roberta Base, they are more likely to be correct for our dataset rather than the Vader results. These results show there is a strong negative correlation in the sentiment around gen ai in games within the dataset.

Part B.3 - Topic Analysis

Now that we understand the sentiment, we can apply topic analysis using Latent Dirichlet allocation to determine the different subtopics within the dataset. These results are shown in the intertopic distance map and the word map below.



The intertopic distance map and word map indicates that the dataset contains several moderately distinct thematic clusters centred around AI, gaming, and generative content. Topic 1 dominates the discussion space, suggesting that AI-assisted game creation is the

primary theme across the dataset, proving that the dataset we collected is appropriate for solving the ultimate question. Topics 2, 4 and 7 relate closely to practical AI usage and collaborative development, while Topics 3, 6, and 10 capture scepticism, emotional reactions, and criticism towards AI-generated content. The spatial separation between topics suggests that discussion ranges from technical implementation to ethical and emotional responses and reflection of discourse around generative AI in games, proving our findings in the part above.

Conclusion

Overall, using sentiment and topic analysis we were able to determine that there is an overall negative sentiment towards generative artificial intelligence in games on the social media platform, Bluesky. We were also able to get an understanding through graphing the first-degree follow networks, that in Bluesky the main type of users discussing the topic of generative ai in games tend to be intertwined with each other showing ego nodes having a least a few second-degree connections. When it comes to the community around the topic, while a majority of the individuals in our reply graph we generated from our dataset are isolated, mostly connecting to only one node, there is a handful that show that they are actively partaking in the discussion as we that they are connected to multiple nodes, demonstrating that they are open to having discussions from different perspectives.